INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab Session 2: Multiplicative inverse mod n *February 26, 2026*

Please solve the following exercises using C/C++, C#, Java or Python programming language to develop your source code, but use only one of them to do all the exercises.

1. Programming exercises I

Do the following exercises by your own. **You must upload screenshots of your source code and screenshots of your programs running, to Microsoft Teams today at the end of the session.**

1. Design a function that implements the Algorithm 1 to find the multiplicative inverse of a^{-1} mód n.

Algoritmo 1: Multiplicative inverse(n,a)

Input : $n \in \mathbb{Z}$, where $n \geq 2$, $a \in \mathbb{Z}_n$
Output : Multiplicative inverse of a, i.e. a^{-1} mód n

```
a_0 ← n ;
b_0 ← a ;
t_0 ← 0 ;
t ← 1 ;
q ← ⌊a_0 / b_0⌋ ;
r ← a_0 − q b_0 ;
while r > 0 do
    temp ← (t_0 − q t) mód n ;
    t_0 ← t ;
    t ← temp ;
    a_0 ← b_0 ;
    b_0 ← r ;
    q ← ⌊a_0 / b_0⌋ ;
    r ← a_0 − q b_0 ;
end
if b_0 ≠ 1 then
    a has no inverse modulo n
else
    return 0
end
```

2. Design and implement a function that receives as a parameter an integer $n \geq 2$ and generates a random valid key K, i.e. a matrix of size 2×2 to be used with the Hill cipher.

3. Design and implement a function that receives as a parameter an integer $n \geq 2$ and a random valid key K, i.e. a matrix of size 2×2 of the Hill cipher. Then use your implementation of Algorithm 1find K^{-1}

4. Use the functions that you previously implemented to obtain the multiplicative inverses for 5 different pair of values (n, a).

5. Use the functions that you previously implemented to obtain 3 different keys K for Hill cipher and their corresponding K^{-1}

2. Programming exercises II

Do the following exercises in teams of 2 students. Use the alphabet of the printable characters in the ASCII i.e. $A = \{space, '!', \ldots, \tilde{}\ \}$ with codes from 32 to 126. Keep the line breaks of the original plaintext. Then do the following:

1. Design and implement a function to encipher a text file using the Hill cipher. Your function must receive as a parameter the filename of the plaintext, a valid key K for the Hill cipher and store the ciphertext in a textfile. Your function must use modular arithmetic with integers module n, where n is the size of the alphabet.

2. Design and implement a function to decipher a ciphertext enciphered with the Hill cipher. Your function must receive a ciphertext (with characters in the set A), a valid key K for the Hill cipher, and recover the original plaintext. Use the function you implemented to compute K^{-1}. Your function must use modular arithmetic with integers module n, where n is the size of the alphabet.

3. You must test your implementation in pairs as follows :

   a) One student of the team, must generate at random a valid key $K$for the Hill cipher considering that the alphabet is the printable ASCII. Then he/she must encipher a text file, for example the source code of any program, using the generated key K. **The filename of the plaintext and the ciphertext, and the key must be given by the user. You must not need to write the filename or the key directly into the source code.** Once he/she obtained the ciphertext, the student must share the key K and the textfile of the ciphertext with his/her partner.

   b) The other student must use K and the textfile of the ciphertext C to recover the plaintext M. **The filename of the ciphertext, the recovered plaintext and the key must be given by the user. You must not need to write the filename or the key directly into the source code.**. Your program must print the value of K^{-1} mód n.

3. Questions

Answer the following questions and write the answers in your report

1. Suppose you are an adversary who is doing the cryptanalysis for Hill cipher. Somehow you have a plaintext M and the corresponding ciphertext C. Is it possible to recover the key used to encipher

M with this information? Explain how can you recover the key. This attack will not be alwways be succesful explain why.

2. Using your implementation of Hill cipher, propose a short (4 characters) plaintext M, obtain the ciphertext C. Do the calculations by hand to recover K.

3. Propose a pair of plaintext M and the corresponding ciphertext C where you will be unable to recover K.

4. Write down the pseudocode of an algorithm that receives as input a plaintext M and the corresponding ciphertext C using Hill cipher. The output must be K. Your algorithm must consider the cases where the attack is not succesful.

4. Products

Every student must write her/his own report, containing at least:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.

2. List 10 keys K obtained by your key generation function and the corresponding K^{-1}.

3. Give 5 pairs of plaintext/ciphertext you used to test your enciphering function.

4. Answers to the questions of Section 3.